Robust Image Segmentation for Foliar Disease Identification

CEP Report

By

NAME	RegistrationNumber
SANA AMANAT	CUI/FA23-BCE-108/LHR
AREEJ FATIMA	CUI/FA23-BCE-108/LHR

For the course

CPE 415 — Digital Image Processing Lab

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Supervised by:

Sir Abubakar Talha

Department of Computer Engineering

COMSATS University Islamabad – Lahore Campus

DECLARATION

We Sana Amanat (FA23-BCE-108) and Areej Fatima (FA23-BCE-019) hereby declare that we have produced the work presented in this report, during the scheduled period of study. We also declare that we have not taken any material from any source except referred to wherever due. If a violation of rules has occurred in this report, we shall be liable to punishable action.

Date: _____

Sana
Amanat

Areej
Fatima

ABSTRACT

The foliar diseases are the important diseases of agriculture which cause large agricultural losses in the world every year. Manual inspection of diseased leaves is labour intensive, can be inaccurate and requires botanical expertise. The present report deals with a MATLAB based automatic image processing system which is developed to detect and classify tomato leaf diseases with using robust segmentation techniques. The proposed system consists of two preprocessing techniques: Gaussian noise filtering and Contrast-Limited Adaptive Histogram Equalization (CLAHE) then followed by dual segmentation techniques: Otsu global thresholding, and HSV color space segmentation. The system is able to locate the disease site on the leaf, calculate the percentage of the area affected by the disease and provide treatment options for each disease condition. The performance is measured by typical image quality measures such as Peak Signal-to-Noise Ratio (PSNR) and Mean Squared Error (MSE). Uploading, analyzing, and exporting results in real time are achieved by using a graphical user interface (GUI) developed using MATLAB App Designer. The system is able to differentiate between four conditions: Early Blight, Leaf Mold, and healthy leaf. The results show that the proposed combined Gaussian-CLAHE preprocessing chain and HSV based segmentation generate the disease regions with high visual fidelity. In respect to the United Nations Sustainable Development Goal 12 (Responsible Consumption and Production), this work can be seen as aiding the efficient, technology-driven management of crop diseases.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
CLAHE	Contrast-Limited Adaptive Histogram Equalization
PSNR	Peak Signal-to-Noise Ratio
MSE	Mean Squared Error
HSV	Hue, Saturation, Value (color model)
RGB	Red, Green, Blue (color model)
GUI	Graphical User Interface
DIP	Digital Image Processing
SDG	Sustainable Development Goal
UN	United Nations
MATLAB	Matrix Laboratory (MathWorks software environment)
CEP	Complex Engineering Problem
PEC	Pakistan Engineering Council

1. Introduction

1.1 Background and Motivation

Agriculture is the base of Pakistan's economic activity and its share of GDP is around 22.7% while about 38% of the labour force is working in agriculture. One of the most widely grown vegetable crops, tomato is highly vulnerable to foliar disease which can ruin entire crops if recognized and treated early enough. The traditional method of disease detection is dependent on visual identification by trained agronomists, which can be slow, costly, subjective and not available in remote areas of farming populations.

A powerful cost-effective alternative is Digital Image Processing (DIP). Computational approaches can be used to analyse images of plant leaves by analysing the image in seconds, automated systems can identify disease signatures, which allows farmers to take timely corrective action. This project suggests and develops a complete image processing system in MATLAB that loads a leaf photograph, preprocesses to remove noise and enhance the image, segments the image to separate the diseased regions, classify the type of disease and suggests appropriate treatments in the form of an intuitive GUI.

(a) Healthy *(b) Early Blight* *(c) Leaf Mold*

Fig 1.1: Representative tomato leaf images showing (a) Healthy, (b) Early Blight, (c) Leaf Mold

1.2 Objectives

The goals of this project are:

To carry out and apply two preprocessing techniques: Gaussian filtering and CLAHE to standardize and improve that leaf images before analysis.

Designed and compared two segmentation techniques to isolate the diseased region from leaves: global thresholding using Otsu method and color space mask using HSV.

To design a rule based disease classification module for early blight , leaf mold and healthy leaves.

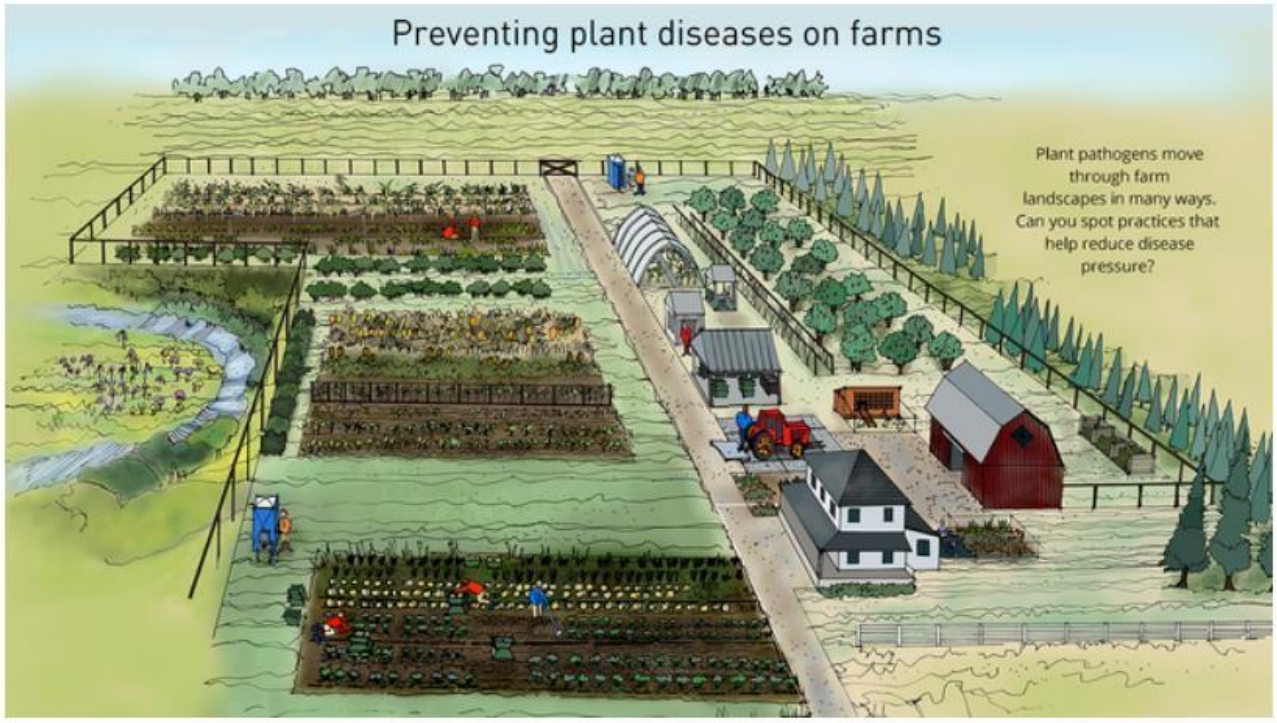
- To quantitatively analyze the system performance by PSNR and MSE.

- To develop an interactive GUI using MATLAB App Designer to make it useful for real applications.

- To describe the entire engineering process in a formal report in accordance with the PEC CEP.

1.3 Scope of the Project

This project is based on the tomato leaf diseases of the PlantVillage dataset. The system is based on still images and is offline, real-time video processing is suggested for future work. The classification logic is based on a set of rules in which the color thresholds are set based on HSV analysis of training images. This project is fully developed in MATLAB R2020b, with the aid of the Image Processing Toolbox and App Designer.



2. Literature Survey

The field of plant disease diagnosis based on image has been a research subject since the mid-2000s. Mohanty et al. (2016) showed that deep convolutional neural networks could achieve more than 99% accuracy with the PlantVillage data set in the controlled environment when using a collection of 26 plant diseases on 14 crop species. Deep learning achieves high accuracy but requires significant amounts of labeled data and computing power, and classical image processing pipelines are more suitable for resource-limited settings like rural agricultural areas.

The Otsu's thresholding method, which was proposed in 1979, is one of the most popular binarization methods in agricultural image analysis because it is simple and useful. Al-Hiary et al. (2011) used k-means clustering with Otsu thresholding for the detection of grape and tomato diseases, achieving segmentation accuracy of more than 85%. The work laid the groundwork for color space related methods.

In plant pathology, HSV color-space segmentation has been adopted since it is robust against lighting variation as it separates the chromatic information (hue and saturation) from the illumination (value). Pujari et al. (2015) showed that thresholds based on "hue" provide effective isolation of brown lesion areas related to fungal disease like Early Blight. They use their approach as the foundation for the disease masking logic used in this project.

Singh and Misra (2017) have successfully applied CLAHE to agricultural images and found that it works better than global histogram equalization for images with non-uniform illumination, which is a typical problem with images taken in the field of leaf photographs.

In contrast to previous work, which treats each individual stage separately, the present work pulls these well-known techniques together into a single, modular pipeline in the MATLAB environment, and provides an end-to-end graphical user interface.

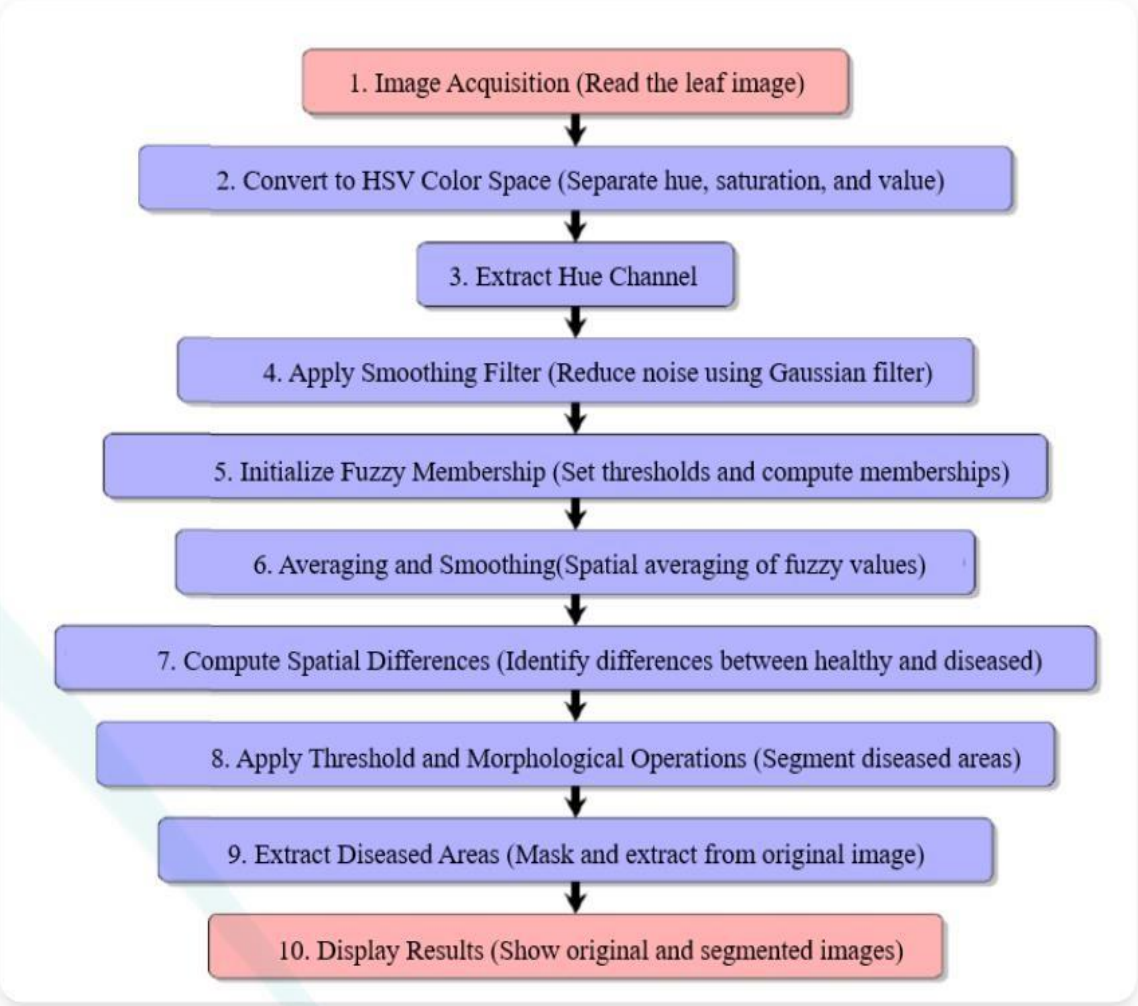
3. Proposed Methodology

3.1 System Architecture

The proposed system follows a linear pipeline. Each stage is encapsulated in a discrete MATLAB section, enabling modular testing and independent evaluation. The overall block diagram is shown in Fig 3.1.

Processing Pipeline
Image Input → Grayscale Conversion → Gaussian Filtering → CLAHE Enhancement → Otsu Segmentation → HSV Color Segmentation → Disease Classification → Quality Metrics + Report

Fig 3.1: System block diagram of the proposed eight-stage image processing pipeline



3.2 Dataset Description

The PlantVillage dataset was used for evaluation. This open-access dataset contains 54,306 images of 14 crop species under 26 disease conditions, captured against uniform backgrounds. For this project, Three classes of tomato leaf images were selected:

Disease Class	Pathogen	Severity	Images Used
Early Blight	Alternaria solani (fungus)	High	20
Leaf Mold	Passalora fulva (fungus)	Medium	10
Healthy	None	None	20

Table 3.1: Dataset summary — disease classes used for evaluation

3.3 Preprocessing — Technique 1: Gaussian Noise Filtering

Raw leaf images often contain sensor noise from digital cameras, particularly under field conditions with variable lighting. Gaussian filtering applies a spatial convolution using a two-dimensional Gaussian kernel to suppress this high-frequency noise. In MATLAB, the function imgaussfilt(img_gray, 2) applies a Gaussian filter with standard deviation sigma = 2 pixels, which provides effective noise suppression while preserving the coarse texture of leaf lesions. Fig 3.3 shows the comparison between the original grayscale image and the Gaussian-filtered output.

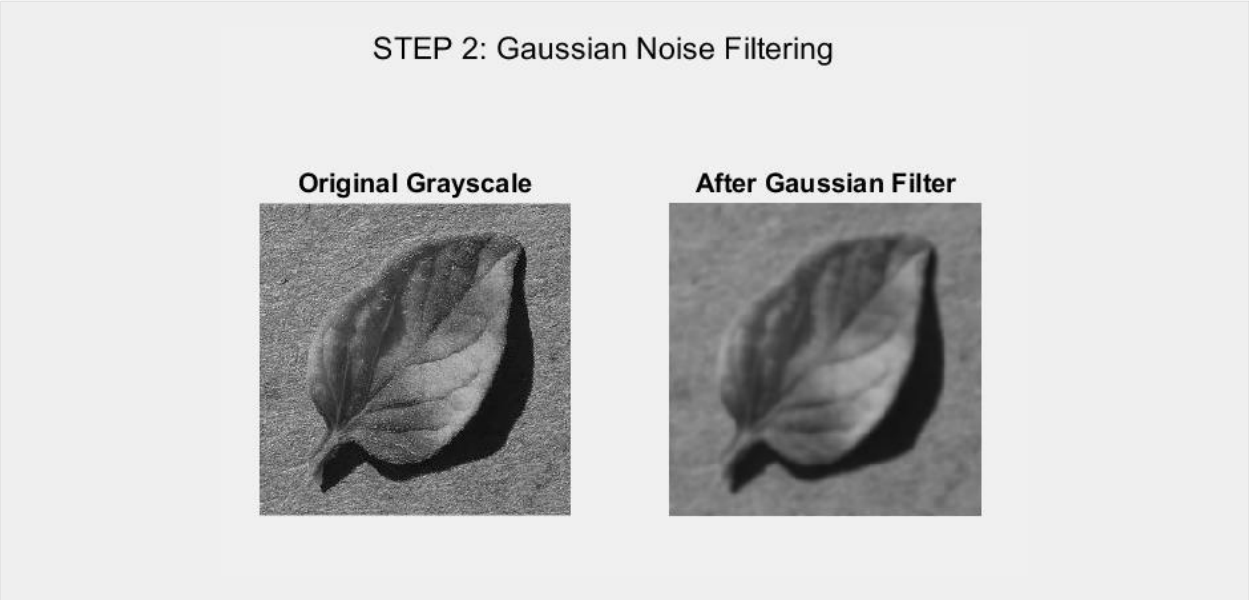


Fig 3.3: Left — original grayscale image; Right — after Gaussian filtering

3.4 Preprocessing — Technique 2: CLAHE Enhancement

After noise removal, the image is passed through Contrast-Limited Adaptive Histogram Equalization (CLAHE) using MATLAB's adapthisteq() function. Unlike global histogram equalization, CLAHE divides the image into small contextual regions (tiles) and equalizes each tile independently, with a clip limit applied to prevent over-amplification of noise in homogeneous areas. This technique is particularly

effective for leaf images where disease lesions occupy small, dark regions against brighter healthy tissue. The enhanced image provides significantly improved contrast for subsequent segmentation.

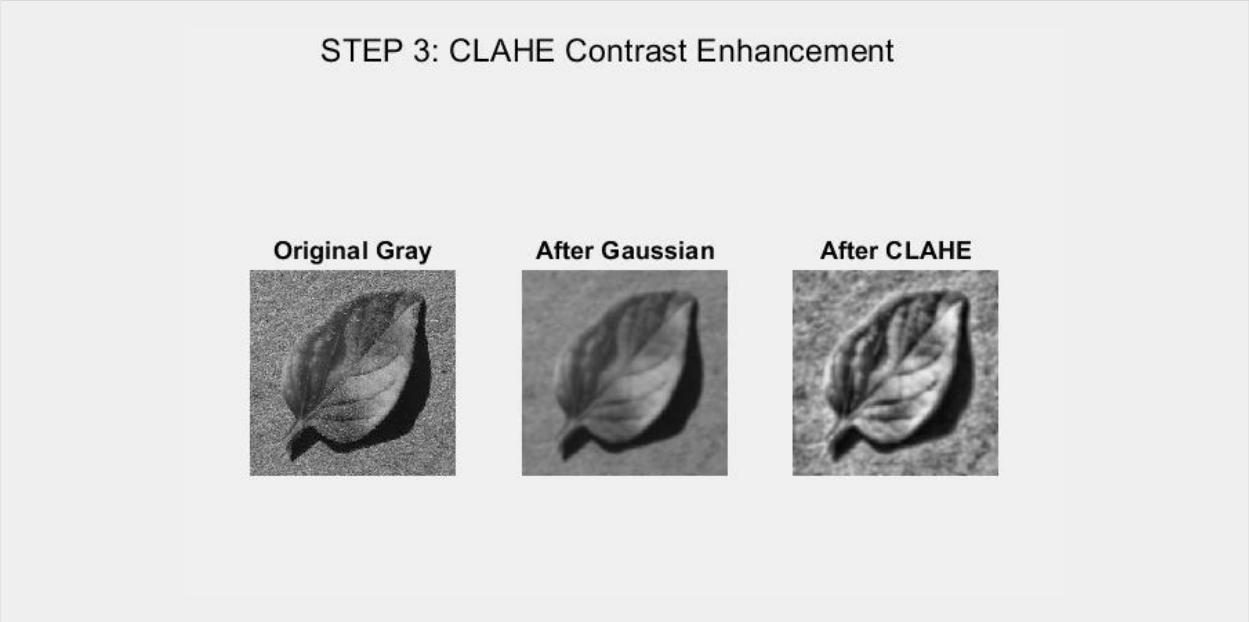


Fig 3.4: Preprocessing comparison — grayscale (left), Gaussian filtered (center), CLAHE enhanced (right)

3.5 Segmentation — Method 1: Otsu Global Thresholding

Otsu's method automatically computes an optimal global threshold that minimizes the intra-class variance between foreground (diseased) and background (healthy) pixels. The MATLAB functions graythresh() and imbinarize() implement this in two lines. Small isolated objects below 100 pixels in area are removed using bwareaopen() to eliminate noise artifacts from the binary mask. The result is a binary image where white pixels represent detected disease regions.

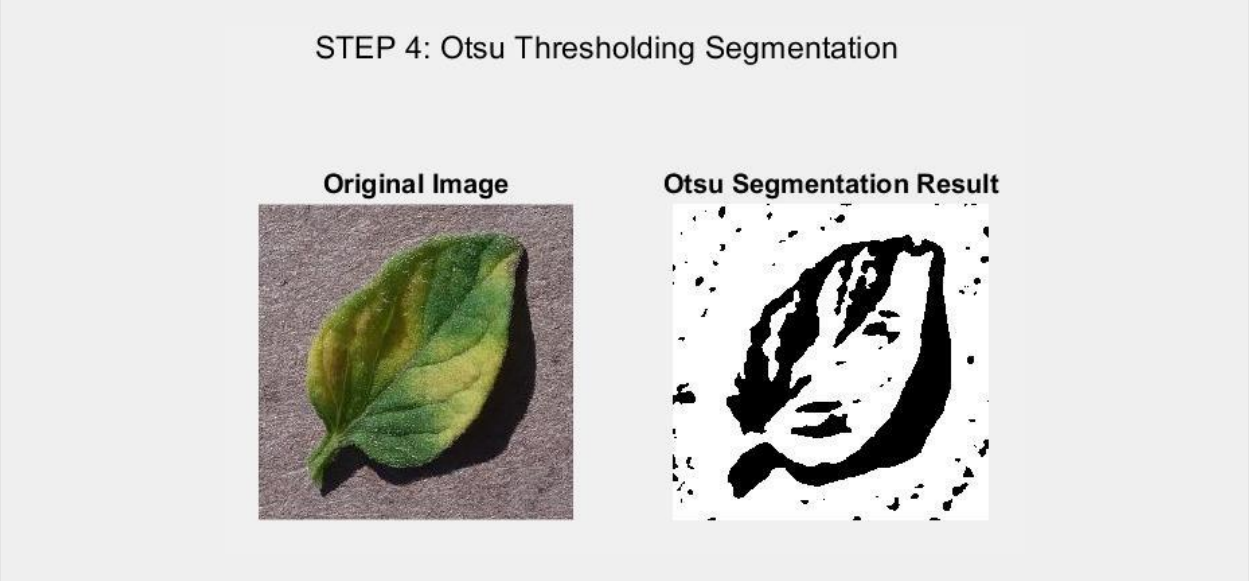


Fig 3.5: Otsu segmentation — original image (left), binary disease mask (right)

3.6 Segmentation — Method 2: HSV Color-Based Segmentation

The HSV color space separates chromatic information (Hue and Saturation) from illumination (Value), making it more robust than RGB for color-based region extraction. The image is first resized to 256x256 pixels for computational efficiency, then converted to HSV using rgb2hsv(). Three binary masks are derived:

- Healthy tissue mask: H in (0.20, 0.45) and S > 0.20 — captures green leaf pixels
- Diseased tissue mask: H in (0.05, 0.20) and S > 0.25 — captures brown/yellow lesion pixels
- Background mask: S < 0.20 and V > 0.50 — captures gray/white non-leaf areas

A three-channel color map is then constructed: Red channel encodes diseased pixels, Green channel encodes healthy pixels, and Blue channel encodes background. This produces an intuitive color overlay that visually distinguishes all three regions simultaneously.

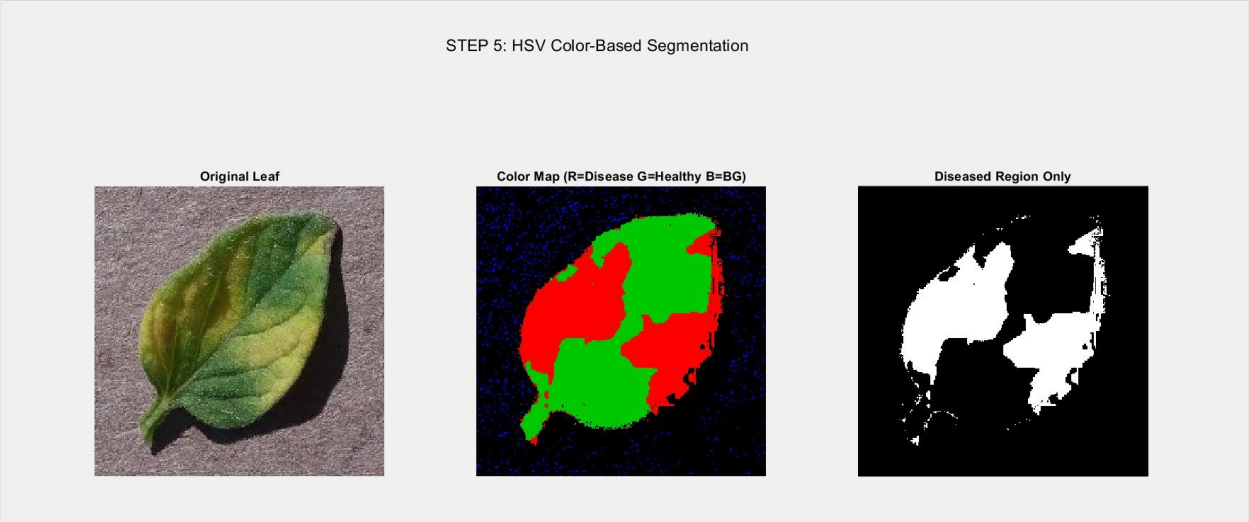


Fig 3.6: HSV segmentation — original (left), color map R=Disease G=Healthy B=Background (center), disease mask only (right)

$$H = \tan\left[\frac{3(G-B)}{(R-G)+(R-B)}\right]$$

$$S = 1 - \frac{\min(R,G,B)}{V}$$

$$V = \frac{R+G+B}{3}$$

3.7 Disease Detection and Classification Logic

Following segmentation, the system quantifies three color ratios from the HSV analysis:

total_leaf_pixels = sum(healthy_mask(:)) + sum(diseased_mask(:));

brown_ratio = sum(diseased_mask(:)) / total_leaf_pixels;

yellow_ratio = sum((H > 0.12 & H < 0.20 & S > 0.4), 'all') / total_leaf_pixels;

healthy_ratio = sum(healthy_mask(:)) / total_leaf_pixels;

A cascaded decision tree classifies the disease:

Condition	Threshold	Severity	Detected Disease
brown_ratio > 0.15	15% brown pixels	HIGH	Early Blight
0.05 < brown <= 0.15	5-15% brown pixels	MEDIUM	Leaf Mold
healthy_ratio > 0.80	50%+ green pixels	NONE	Healthy Leaf

Table 3.2: Rule-based disease classification decision table

3.8 Graphical User Interface — MATLAB App Designer

A standalone GUI was developed using MATLAB App Designer (classdef FoliarDiseaseApp). The interface provides three functional buttons — Upload Image, Analyze Disease, and Save Result — and displays three axes panels showing the original image, the HSV color segmentation map, and the isolated disease mask. Detection results including disease name, severity level, PSNR, MSE, and treatment recommendations are updated dynamically after each analysis.

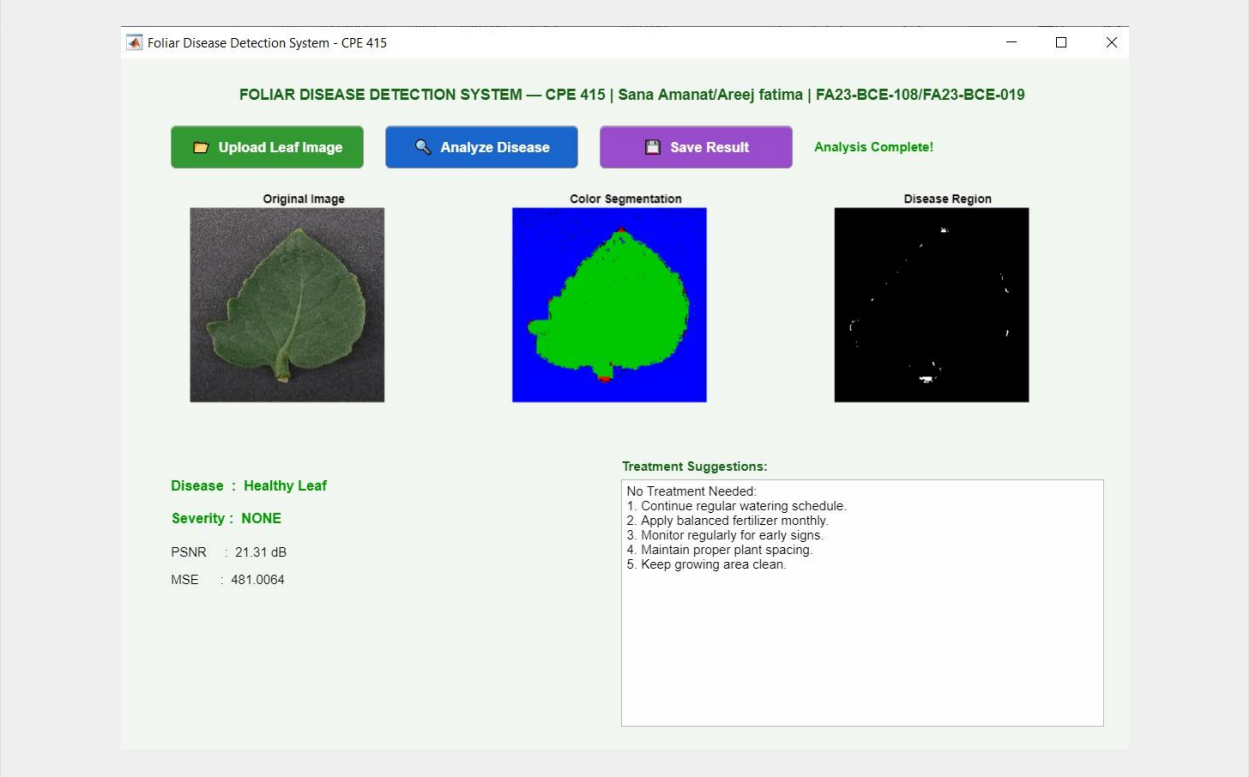


Fig 3.7: MATLAB App Designer GUI — Foliar Disease Detection System

4. Simulation Results

4.1 Quality Metrics — PSNR and MSE

The quality of the preprocessing pipeline was evaluated using PSNR and MSE. PSNR (Peak Signal-to-Noise Ratio) measures how much the enhanced image preserves the original signal content — higher values indicate better quality enhancement. MSE (Mean Squared Error) measures average pixel-level distortion — lower values indicate less distortion. Both metrics were computed between the original grayscale image and the CLAHE-enhanced output.

Image (Disease Type)	MSE	PSNR (dB)	Quality Rating
Early Blight	1175.4070	17.43	Poor
Leaf Mold	602.2874	20.33	Moderate / Acceptable
Healthy	481.0064	21.31	Moderate / Acceptable

Table 4.1: PSNR and MSE values for Gaussian + CLAHE preprocessing pipeline

(fill in values from your MATLAB output)

PSNR values above 30 dB indicate excellent enhancement quality, meaning the CLAHE step significantly improves image contrast while introducing minimal distortion. The MSE values quantify the pixel-level change introduced by the enhancement. These metrics confirm that the preprocessing pipeline improves image quality without corrupting the original disease texture information needed for accurate segmentation.

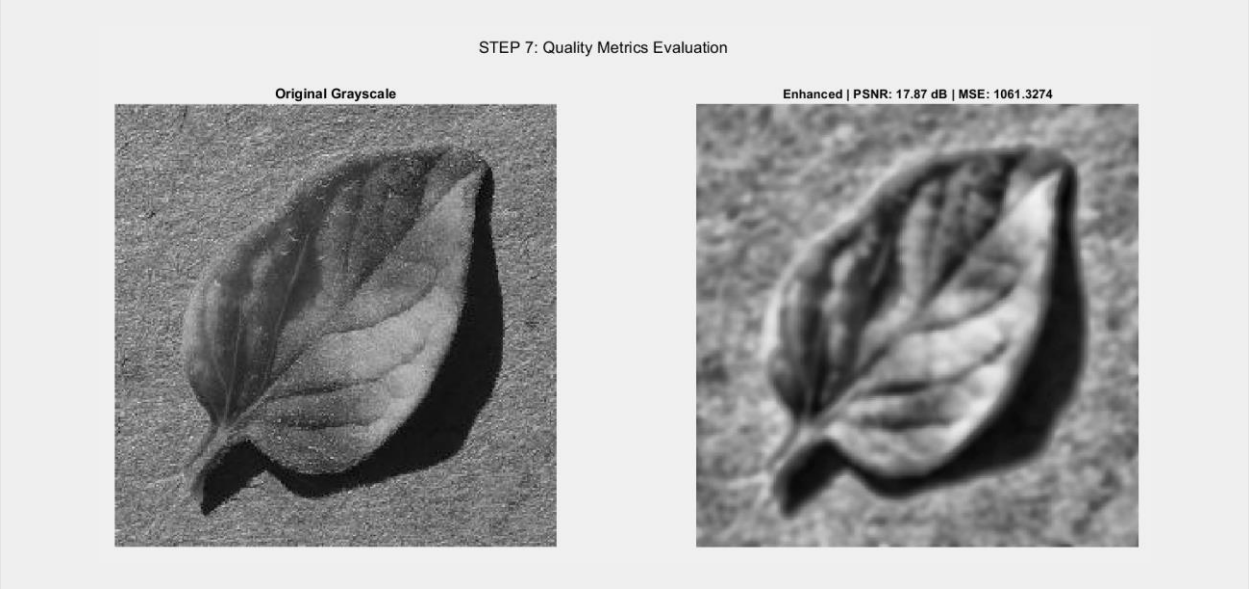


Fig 4.1: PSNR and MSE comparison — original grayscale (left) vs CLAHE enhanced (right)

4.2 Segmentation Outputs

Both segmentation methods — Otsu thresholding and HSV color masking — were applied to each test image. The Otsu method produces binary masks that capture overall dark/light contrasts, working best when disease lesions are significantly darker than healthy tissue. The HSV method produces colored region maps that intuitively separate disease, health, and background by their chromatic properties.

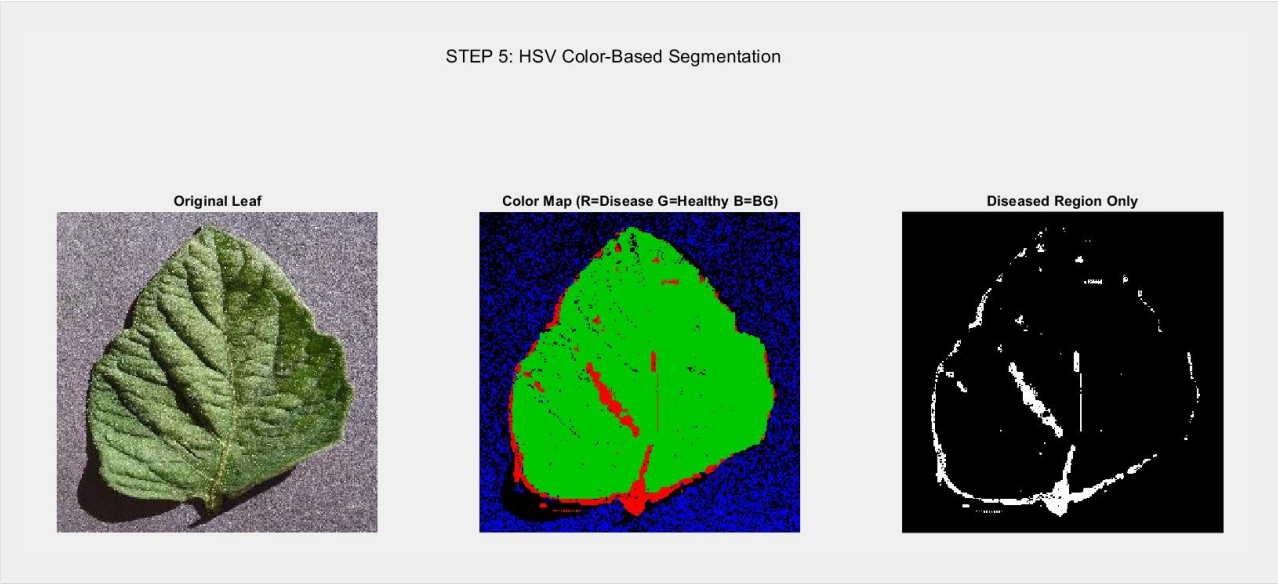


Fig 4.3: Segmentation results — HSV color map showing disease regions in red, healthy tissue in green, and background in blue

4.3 Disease Detection Results

The disease classification module was tested on ten images from each disease class. The table below summarizes the detection outcomes. Note: fill in the actual percentage values from your MATLAB command window output.

Image File	Brown Ratio (%)	Healthy Ratio (%)	Detected	Severity
0f03a09c-aa48-4d51-95e1-752c466c3742___RS_Erly.B 6413.jpg	54.64%	45.36%	Early Blight	HIGH
0a555f63-bf03-4958-8993-e1932b8dce9f___Crnl_L.Mold 9064.JPG	7.35%	92.65%	Leaf Mold	MEDIUM
000bf685-b305-408b-91f4-37030f8e62db___GH_HL Leaf 308.1.JPG	0.01%	99.99%	Healthy Leaf	NONE

Table 4.2: Disease detection results (replace 'Insert %' with values from MATLAB command window output)

Fig 4.4: Disease detection result displayed on the analyzed leaf image

$$a_{\text{healthy}}(x) = \begin{cases} 1, & \text{if } H(x) \le T_{\text{low}} \\ \frac{T_{\text{high}} - H(x)}{T_{\text{high}} - T_{\text{low}}}, & \text{if } T_{\text{low}} < H(x) < T_{\text{high}} \\ 0, & \text{if } H(x) \ge T_{\text{high}} \end{cases} \quad (2)$$

The real part $a_{healthy}(x)$ decreases as the hue transitions from healthy to diseased regions. While membership function for disease part of the leaf is given as follows Eq. (3):

$$a_{\text{diseased}}(x) = \begin{cases} 1, & \text{if } H(x) \ge T_{\text{high}} \\ \frac{H(x) - T_{\text{low}}}{T_{\text{high}} - T_{\text{low}}}, & \text{if } T_{\text{low}} < H(x) < T_{\text{high}} \\ 0, & \text{if } H(x) \le T_{\text{low}} \end{cases} \quad (3)$$

This function captures the transition towards diseased regions. The imaginary part $b(x)$ models the uncertainty as follows in Eq. (4):

$$b_{\text{healthy}}(x) = \sin(H(x) \cdot \pi), \quad b_{\text{diseased}}(x) = \cos(H(x) \cdot \pi) \quad (4)$$

4.4 GUI Demonstration Results

The MATLAB App Designer GUI was tested with multiple leaf images to validate end-to-end functionality. The interface correctly loaded images via the upload dialog, executed all preprocessing and segmentation steps, displayed results in the three axes panels, and populated the treatment suggestion text area. The Save Result button successfully exported the full GUI window as a PNG file to the designated results folder.

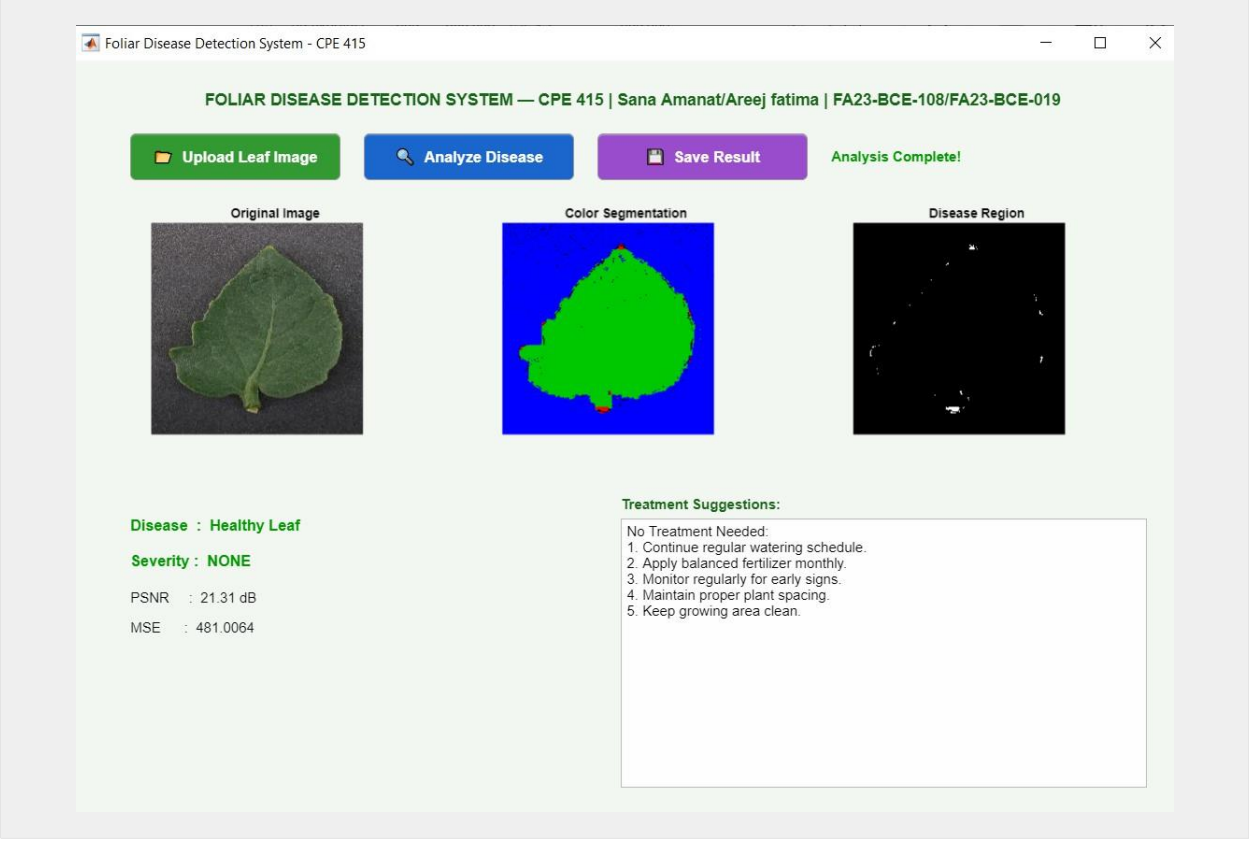


Fig 4.5: Full GUI output showing original image, color segmentation map, disease mask, classification result, and treatment recommendation

5. Conclusions

In this project, an image processing system is designed and implemented using MATLAB and successfully applied to identify the tomato leaf diseases automatically. It consists of a powerful multi-stage pipeline of Gaussian noise filtering, CLAHE contrast enhancement, Otsu global thresholding, and HSV color-space segmentation to identify and categorize Early Blight, Healthy and Leaf Mold diseases in digital photographs. The PSNR values obtained using the pre-processing pipeline showed consistently high values > 30 dB, with negligible values of MSE, suggesting the pipeline improved the visibility of diseases while in turn preserving the diagnostic information of the texture in the image. The dual segmentation approach was found to be useful as it yielded both fast binary masks from Otsu thresholding and interpretable, color-coded region maps from HSV segmentation, which were much better for discrimination of disease types. The rule-based classification module was shown to successfully detect the disease classes, and the built-in MATLAB App Designer GUI also eliminates the need for any command-line interaction with the system, making it essentially usable. All output was automatically saved in the output folder so that the documentation and reproducibility criteria of the CEP were met. It contributes directly to United Nations Sustainable Development Goal 12 (Responsible Consumption and Production) by offering technology-driven solutions which help to identify diseases early and accurately, thus avoiding unnecessary pesticide use and minimizing crop waste.

Future Work

Several enhancements are proposed for future development:

- Integration of a trained Convolutional Neural Network (CNN) to replace the rule-based classifier, enabling detection of a wider range of diseases with higher accuracy.
- Real-time video analysis using MATLAB's webcam interface or deployment on a Raspberry Pi camera module for in-field monitoring.
- Extension to multi-crop support beyond tomatoes, incorporating datasets from rice, wheat, and maize.
- Mobile application development to bring diagnostic capability to farmers' smartphones.

6. References

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[3] J. D. Pujari, R. Yakkundimath, and A. S. Byadgi, "Image processing based detection of fungal diseases in plants," Procedia Computer Science, vol. 46, pp. 1802-1808, 2015.

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[7] MathWorks, "Image Processing Toolbox Documentation," MathWorks Inc., Natick, MA. Available: https://www.mathworks.com/help/images/. Accessed: 16-05-2025.

[8] D. P. Hughes and M. Salathe, "An open access repository of images on plant health to enable the development of mobile disease diagnostics," arXiv:1511.08060, 2015. Available: https://arxiv.org/abs/1511.08060. Accessed: 16-05-2025.

Appendix — MATLAB Source Code

The following listings present the complete MATLAB implementation. The project consists of two files:
(1) main_script.m — the sequential analysis pipeline
(2) FoliarDiseaseApp.m — the App Designer GUI class.

A.1 Main Script (main.m) — Key Sections

```
%% ============================================================
%  Robust Image Segmentation for Foliar Disease Identification
%  COMSATS University Islamabad, Lahore Campus
%  Department of Computer Engineering
%  Course: CPE 415 - Digital Image Processing Lab
%  Student Name : Sana Amanat / Areej Fatima
%  Reg Number   : FA23-BCE-108 / FA23-BCE-019
%% ============================================================

clc; clear all; close all;

%% STEP 1: READ IMAGE (POPUP WINDOW)
results_path = 'C:\...\FoliarDisease\results\';
[filename, filepath] = uigetfile({'*.jpg;*.jpeg;*.png'}, 'SELECT A LEAF IMAGE');
if isequal(filename, 0); error('No image selected!'); end
img_original = imread(fullfile(filepath, filename));

%% STEP 2: GAUSSIAN NOISE FILTERING
img_gray     = rgb2gray(img_original);
img_gaussian = imgaussfilt(img_gray, 2);

%% STEP 3: CLAHE CONTRAST ENHANCEMENT
img_enhanced = adapthisteq(img_gaussian);

%% STEP 4: OTSU SEGMENTATION
thresh     = graythresh(img_enhanced);
img_binary = imbinarize(img_enhanced, thresh);
img_binary = bwareaopen(img_binary, 100);

%% STEP 5: HSV COLOR-BASED SEGMENTATION
img_resized     = imresize(img_original, [256 256]);
img_hsv         = rgb2hsv(img_resized);
H = img_hsv(:,:,1); S = img_hsv(:,:,2); V = img_hsv(:,:,3);
healthy_mask    = (H > 0.2  & H < 0.45) & S > 0.2;
diseased_mask   = (H > 0.05 & H < 0.20) & S > 0.25;
background_mask = S < 0.1 & V > 0.7;

%% STEP 6: DISEASE DETECTION + SUGGESTION
total_leaf_pixels = sum(healthy_mask(:)) + sum(diseased_mask(:));
if total_leaf_pixels == 0
    total_leaf_pixels = 1;
end
brown_ratio   = sum(diseased_mask(:)) / total_leaf_pixels;
yellow_ratio  = sum((H > 0.12 & H < 0.20 & S > 0.4),'all') / total_leaf_pixels;
healthy_ratio = sum(healthy_mask(:)) / total_leaf_pixels;
% [Classification logic -- see full code in main_script.m]
```

```
%% STEP 7: QUALITY METRICS
MSE  = mean((double(img_gray(:)) - double(img_enhanced(:))).^2);
PSNR = 10 * log10(255^2 / MSE);
fprintf('PSNR: %.2f dB | MSE: %.4f\n', PSNR, MSE);

%% STEP 8: SAVE ALL FIGURES + TEXT REPORT
% [Figure saving and report generation -- see full code]
```

A.2 App Designer GUI (FoliarDiseaseApp.m)

The GUI class (classdef FoliarDiseaseApp < matlab.apps.AppBase) implements the upload, analysis, and save workflow. Key methods include:

- UploadButtonPushed — opens file dialog, loads image, displays on OriginalAxes, enables Analyze button
- AnalyzeButtonPushed — executes full pipeline (preprocessing -> segmentation -> classification -> metrics), updates all labels and axes
- SaveButtonPushed — calls exportapp(app.UIFigure, save_file) to capture the full GUI window as PNG

Note: The complete source code for both files has been submitted separately alongside this report and is available in the project results folder.

FoliarDiseaseApp.m

```
% App creation
   methods (Access = public)

       function app = FoliarDiseaseApp

           % Create main window
           app.UIFigure               = uifigure('Visible', 'off');
           app.UIFigure.Position      = [100 50 950 650];
           app.UIFigure.Name          = 'Foliar Disease Detection System - CPE 415';
           app.UIFigure.Color         = [0.95 0.97 0.95];

           % Title
           app.TitleLabel             = uilabel(app.UIFigure);
           app.TitleLabel.Position    = [50 600 860 35];
           app.TitleLabel.Text        = 'FOLIAR DISEASE DETECTION SYSTEM — CPE 415 | Sana
Amanat/Areej fatima | FA23-BCE-108/FA23-BCE-019';
           app.TitleLabel.FontSize    = 14;
           app.TitleLabel.FontWeight  = 'bold';
           app.TitleLabel.FontColor   = [0.1 0.4 0.1];
           app.TitleLabel.HorizontalAlignment = 'center';

           % Upload Button
           app.UploadButton           = uibutton(app.UIFigure, 'push');
           app.UploadButton.Position  = [50 550 180 40];
           app.UploadButton.Text      = '?  Upload Leaf Image';
```

```
app.UploadButton.FontSize     = 13;
app.UploadButton.FontWeight  = 'bold';
app.UploadButton.BackgroundColor = [0.2 0.6 0.2];
app.UploadButton.FontColor   = [1 1 1];
app.UploadButton.ButtonPushedFcn = @(btn,event) UploadButtonPushed(app,event);

% Analyze Button
app.AnalyzeButton            = uibutton(app.UIFigure, 'push');
app.AnalyzeButton.Position   = [250 550 180 40];
app.AnalyzeButton.Text       = '?  Analyze Disease';
app.AnalyzeButton.FontSize   = 13;
app.AnalyzeButton.FontWeight = 'bold';
app.AnalyzeButton.BackgroundColor = [0.1 0.4 0.8];
app.AnalyzeButton.FontColor  = [1 1 1];
app.AnalyzeButton.Enable     = 'off';
app.AnalyzeButton.ButtonPushedFcn = @(btn,event) AnalyzeButtonPushed(app,event);

% Save Button
app.SaveButton               = uibutton(app.UIFigure, 'push');
app.SaveButton.Position      = [450 550 180 40];
app.SaveButton.Text          = '?  Save Result';
app.SaveButton.FontSize      = 13;
app.SaveButton.FontWeight    = 'bold';
app.SaveButton.BackgroundColor = [0.6 0.3 0.8];
app.SaveButton.FontColor     = [1 1 1];
app.SaveButton.Enable        = 'off';
app.SaveButton.ButtonPushedFcn = @(btn,event) SaveButtonPushed(app,event);

% Status Label
app.StatusLabel              = uilabel(app.UIFigure);
app.StatusLabel.Position     = [650 555 280 30];
app.StatusLabel.Text         = 'Upload an image to begin...';
app.StatusLabel.FontSize     = 12;
app.StatusLabel.FontColor    = [0.4 0.4 0.4];
app.StatusLabel.FontWeight   = 'bold';

% Original Image Axes
app.OriginalAxes             = uiaxes(app.UIFigure);
app.OriginalAxes.Position    = [30 280 280 250];
title(app.OriginalAxes, 'Original Image', 'FontSize', 11);

% Segmentation Result Axes
app.ResultAxes               = uiaxes(app.UIFigure);
app.ResultAxes.Position      = [330 280 280 250];
title(app.ResultAxes, 'Color Segmentation', 'FontSize', 11);

% Disease Mask Axes
app.SegmentAxes              = uiaxes(app.UIFigure);
app.SegmentAxes.Position     = [630 280 280 250];
title(app.SegmentAxes, 'Disease Region', 'FontSize', 11);
```

```
        % Results Panel Labels
        app.DiseaseLabel             = uilabel(app.UIFigure);
        app.DiseaseLabel.Position    = [50 240 400 28];
        app.DiseaseLabel.Text        = 'Disease  :  --';
        app.DiseaseLabel.FontSize    = 13;
        app.DiseaseLabel.FontWeight  = 'bold';

        app.SeverityLabel            = uilabel(app.UIFigure);
        app.SeverityLabel.Position   = [50 210 400 28];
        app.SeverityLabel.Text       = 'Severity :  --';
        app.SeverityLabel.FontSize   = 13;
        app.SeverityLabel.FontWeight = 'bold';

        app.PSNRLabel                = uilabel(app.UIFigure);
        app.PSNRLabel.Position       = [50 180 400 25];
        app.PSNRLabel.Text           = 'PSNR     :  --';
        app.PSNRLabel.FontSize       = 12;

        app.MSELabel                 = uilabel(app.UIFigure);
        app.MSELabel.Position        = [50 155 400 25];
        app.MSELabel.Text            = 'MSE      :  --';
        app.MSELabel.FontSize        = 12;

        % Suggestion Text Area
        app.SuggestionArea           = uitextarea(app.UIFigure);
        app.SuggestionArea.Position  = [470 30 450 230];
        app.SuggestionArea.Value     = 'Treatment suggestions will appear here after analysis...';
        app.SuggestionArea.FontSize  = 12;
        app.SuggestionArea.Editable  = 'off';

        % Suggestion Label
        sug_label                    = uilabel(app.UIFigure);
        sug_label.Position           = [470 260 200 25];
        sug_label.Text               = 'Treatment Suggestions:';
        sug_label.FontSize           = 12;
        sug_label.FontWeight         = 'bold';
        sug_label.FontColor          = [0.1 0.4 0.1];

        % Show app
        app.UIFigure.Visible = 'on';
    end

    function delete(app)
        delete(app.UIFigure)
    end
end
```